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Key Points:

- Tibetan Plateau's dominant forcing on the Northern Hemisphere subtropical high shifts from mechanical in winter to thermal in summer
- Tibetan Plateau uplift drives 2.3° northward shift of the Northern Hemisphere subtropical high and intensifies it in summer
- Tibetan Plateau's mechanical and thermal forcings affect the Northern Hemisphere subtropical high by altering atmospheric circulations

Supporting Information:

Supporting Information may be found in the online version of this article.

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How Mechanical and Thermal Forcing of the Tibetan Plateau Shapes the Northern Hemisphere Subtropical High

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Abstract The subtropical high drives extreme weather and climate in mid-low latitudes. While the summertime western Pacific subtropical high has been extensively studied, the hemisphere-wide influence of the Tibetan Plateau (TP) on subtropical high remains less clear. Multi-model simulations show that TP uplift intensifies and extends the Northern Hemisphere subtropical high northward in summer, while driving an overall intensification and expansion of the western Pacific subtropical high in winter. The mechanical effect of the TP dominates winter variability, whereas summer thermal forcing—peaking with diabatic heating—exerts a primary influence across the hemisphere. These impacts are mediated by atmospheric circulation. The reanalysis data further confirm the important role of TP thermal forcing in the observed variability of the subtropical high. Therefore, TP forcing is essential for understanding and predicting variability of the subtropical high throughout the Northern Hemisphere, beyond the western Pacific sector alone.

Plain Language Summary The subtropical high belt lies between the tropics and mid-latitudes, including some of the driest deserts and the rainiest monsoon areas. Because of this strong contrast, the subtropical high drives highly changeable weather and climate. The Tibetan Plateau (TP), called the “Roof of the World,” can strongly influence the western Pacific subtropical high in summer. In this study, observations and model simulations were used to explore how the TP affects the variability of subtropical high across the Northern Hemisphere. We found that the TP uplift pushes Northern Hemisphere subtropical high northward in summer and expands western Pacific subtropical high northward in other seasons. Our results highlight that the TP primarily influences subtropical high through its mechanical forcing in winter, whereas in summer, its strong thermal forcing becomes the dominant factor.

1. Introduction

The subtropical high is a semi-permanent high-pressure system in the mid-to-lower troposphere. Shaped by land-sea thermal contrasts and topographic forcing, it splits into several regional high-pressure centers, giving rise to arid zones and monsoon regions (W. Li et al., 2012; Sun et al., 2017; Zhu et al., 1986). Its variability strongly affects the position of rain belts (W. Li et al., 2011; Z.-Q. Zhou et al., 2021), extreme heat events (Beniston, 2004; N. Li et al., 2021; M. Luo & Lau, 2017), and tropical cyclone activity (Q. Wu et al., 2020). For example, the westward extension of the western Pacific subtropical high (WPSH) during the 1998 and 2020 Meiyu seasons caused catastrophic flooding in the Yangtze River basin (Jonkman, 2005; Z.-Q. Zhou et al., 2021). Furthermore, variations of the subtropical high on interannual to decadal scales modulate tropical cyclone genesis, frequency, and tracks. Understanding these multi-scale variabilities is crucial for improving regional climate and weather predictions (Bregy et al., 2022; H. Li et al., 2019; C. Wang & Wang, 2019; Q. Wu et al., 2020).

The formation of the subtropical high results from the combined effects of Earth's rotation and diabatic heating, with its meridional and zonal displacements largely determined by the spatial distribution of diabatic heating (G. Wu et al., 2012). Many studies have demonstrated that the nonuniform distribution of diabatic heating is a key factor controlling the position and intensity of the subtropical high (Liu et al., 2001; Rodwell & Hoskins, 2001). In summer, latent heating release associated with the monsoon plays a crucial role in the formation and maintenance of the subtropical high (Rodwell & Hoskins, 2001). In other seasons, particularly in winter, the subtropical high is less directly influenced by monsoonal latent heating and is more strongly shaped by large-scale atmospheric dynamics. Qiu et al. (2023) suggested that the winter subtropical high is mainly regulated by the descending branch of the Hadley circulation and baroclinic instability.

The Tibetan Plateau (TP) affects regional and global climate by altering atmospheric circulation through its mechanical uplift, strong thermal forcing, and complex land-atmosphere interactions (Sun et al., 2021; G. Wu et al., 2007; Xie, Huang, et al., 2023). Of particular interest is its significant influence on the subtropical high. TP uplift can reshape atmospheric circulation globally, influencing the formation and distribution of WPSH (Jiang et al., 2008). Previous studies have demonstrated that anomalous thermal forcing over the TP can modulate WPSH (Duan et al., 2013; Liu et al., 2004; H. Luo et al., 2023; G. Wu & Liu, 2003). One pathway involves the modulation of the South Asian High, as TP thermal forcing induces its eastward extension, enhances descending motion on the western flank of the WPSH, and consequently promotes the westward expansion of the WPSH (Ge et al., 2019; B. Wang et al., 2008; Z. Wang et al., 2014). In addition, strong sensible and latent heating over the TP drives the summer monsoon circulation, thereby affecting regional precipitation patterns and the intensity of the subtropical high (G. Wu & Liu, 2003; G. Wu et al., 2007). For example, springtime heating anomalies over the TP not only weaken the WPSH concurrently but also trigger its delayed strengthening in the following summer, a response mediated by air-sea interactions over the Pacific (Duan et al., 2017).

Building on previous studies that established the TP as a key driver of the WPSH, this study further investigates how the separated mechanical and thermal forcings of the TP affect the spatial position and horizontal extent of the entire Northern Hemisphere subtropical high. After expounding the climatological characteristics of subtropical high using reanalysis data, we analyze the overall, mechanical, and thermal effects of the TP on subtropical high, respectively, using three model simulations, and propose the underlying dynamical mechanisms including Rossby waves, the westerly jet stream, and meridional-vertical circulation cells.

2. Data and Methods

2.1. Reanalysis Data

The ERA5 data set, developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), provides global atmospheric reanalysis data (Hersbach et al., 2020). In this study, data spanning the period from 1940 to 2022 were utilized. The monthly mean 500-hPa geopotential height, together with the stream function calculated from the zonal and meridional wind components, is used to represent the spatial distribution of the subtropical high. In addition, the vertically integrated atmospheric apparent heat source was calculated based on the three-hourly temperature data at pressure levels to quantify the total diabatic heating over the TP (Yanai et al., 1973). To analyze the dynamical mechanisms associated with the TP thermal forcing, the monthly mean geopotential height, meridional wind, pressure-coordinate vertical velocity, and total precipitation were examined.

2.2. Numerical Simulations and Models

To distinguish the overall, mechanical, and thermal effects of the TP, three numerical experiments were designed in this study. The experiments were conducted using version 2.1.3 of the Community Earth System Model (CESM), developed by the National Center for Atmospheric Research (NCAR) and the University Corporation for Atmospheric Research (UCAR) (Danabasoglu et al., 2020). The atmospheric component of CESM employs a finite volume grid with a horizontal resolution of $0.9^\circ \times 1.25^\circ$ (192×288) and 32 vertical levels. These experiments were conducted with prescribed sea surface temperatures and sea ice for 1979–2014. This approach isolates the atmospheric response by preventing long-term oceanic feedbacks, which require multi-century equilibration times following topographic changes (Su et al., 2018; Yang et al., 2020), while retaining active land-atmosphere coupling. The experimental configurations are as follows:

1. Control experiment (CTRL): This simulation follows the historical atmosphere-only experiment of the Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring et al., 2016). It prescribes external forcings, including greenhouse gas concentrations, aerosol emissions, solar irradiance, and land-use changes, for 1979–2014.
2. No-Topography experiment (noTP): Following the Global Monsoon Model Intercomparison Project (GMMIP; T. Zhou et al., 2016), it is derived from the CTRL run but flattens the Asian topography by reducing all surface elevations above 500 m to this base level (22.5°N – 55°N , 35°E – 110°E).
3. No-Sensible-Heating experiment (TPnosh): This experiment retains the CTRL topography but removes the dominant thermal forcing by eliminating sensible heating over the TP (G. Wu et al., 2007). The sensible heating flux is set to zero at all model levels for each integration time step over the Asian topography above

500 m (22.5°N–55°N, 35°E–110°E), a domain consistent with that modified in the noTP experiment. While only sensible heating is explicitly manipulated in the model setup, this single adjustment triggers cascading effects via land-atmosphere coupling and intrinsic atmospheric feedback processes within the model. Consequently, in the TPnosh experiment, changes are not limited to sensible heating alone but also extend to precipitation, latent heating, and other diabatic components. This cascade of responses ultimately alters the total atmospheric thermal forcing over the TP—the primary target of this experimental design—rather than just the sensible heating component (G. Wu et al., 2016; Xie, Huang, et al., 2023).

The distinct impacts of the TP forcing are isolated by comparing the experimental results: the overall effect is represented by CTRL—noTP, the thermal effect by CTRL—TPnosh, and the mechanical effect by TPnosh—noTP. This is justified because the CTRL run includes both mechanical and thermal forcings; the noTP run includes neither; and the TPnosh run retains mechanical forcing while directly modifying the sensible heating, which ultimately leads to changes in the total diabatic heating through feedback processes in the model. In addition, among the GMMIP models, only FGOALS-f3-L (B. He et al., 2020) and FIO-ESM-2-0 (Bao et al., 2020) provide the required monthly output for all three experiments. The FGOALS-f3-L model employs a C96 finite volume grid (180 × 360) with 32 vertical layers, whereas the FIO-ESM-2-0 model adopts a 0.9° × 1.25° finite volume grid (192 × 288) and 26 vertical layers. The GMMIP CTRL experiment is integrated from 1870 to 2014, while its noTP and TPnosh experiments follow the same experimental design as those in our CESM simulations. A three-model mean (CESM, FGOALS-f3-L, FIO-ESM-2-0) was used to examine the impact of the TP. For a signal to be deemed robust, it must be statistically significant in the multi-model mean and exhibit a consistent sign in all three models (see Section 2.4).

2.3. Diagnosis of the Location and Spatial Extent of the Subtropical High

The subtropical high is commonly identified in climate studies and operational monitoring by specific geopotential height contours at 500 hPa, such as the 5,880 gpm line for WPSH in summer. However, this method has notable limitations: (a) geopotential height contours lose their utility in the low latitudes due to the breakdown of geostrophic balance; (b) specific geopotential height values for defining the subtropical high vary across seasons (Lu, 2001), lacking a consistent standard; (c) thermodynamic expansion in a warming atmosphere raises geopotential height, causing these characteristic contours to expand outward even without circulation changes (C. He et al., 2015; L. Wu & Wang, 2015). Since the subtropical high is fundamentally a circulation system, the stream function is employed for diagnosis as it directly represents the horizontal flow, offering a more accurate depiction, particularly in the non-geostrophic low-latitudes.

For the non-divergent component of a two-dimensional flow, a scalar stream function, ψ , is defined such that

$$d\psi = vdx - udy, \quad (1)$$

where u and v are the zonal and meridional wind components, and x and y are the respective zonal and meridional coordinates. The stream function provides a scalar representation of a complex velocity field, which effectively characterizes the large-scale flow structure. Since the stream function ψ lacks a unique solution without additional constraints, the spherical harmonic algorithm in NCL's `uv2sfvpF` function (NCL, 2015) ensures uniqueness by setting its global mean to zero.

2.4. Statistical Method

A two-tailed Student's t -test was employed to assess the statistical significance of differences between the control and sensitivity experiments. The test was applied to the 36-year time series (1979–2014), where each data point represents a seasonal mean for either June–August (JJA) or December–February (DJF). Differences exceeding the 95% confidence level ($p < 0.05$) were deemed statistically significant.

In addition, a composite analysis was employed to investigate the influence of TP thermal forcing on the Northern Hemisphere subtropical high during summer. Years with TP vertically integrated atmospheric apparent heat source anomalies exceeding ± 1 standard deviation were defined as high- and low-forcing years, respectively. The vertically integrated atmospheric apparent heat source is used here to represent the total atmospheric thermal forcing of the TP, including both sensible heating and latent heat release associated with moist processes, consistent with the design of the TPnosh experiment. Composite fields were constructed for each category, and

their differences were used to identify the climatic signature of TP thermal forcing on the subtropical high. The statistical significance of this composite difference was assessed using a two-tailed Student's *t*-test at the 95% confidence level ($p < 0.05$).

2.5. Atmospheric Apparent Heat Source

The calculation of the atmospheric apparent heat source (Q_1) utilizes temperature, wind, vertical velocity, and pressure fields (Yanai et al., 1973). The equation of Q_1 at pressure-level coordinates is given by

$$Q_1 = C_p \left(\frac{p}{p_0} \right)^\kappa \left[\frac{\partial \theta}{\partial t} + \vec{V} \cdot \nabla_p \theta + \omega \frac{\partial \theta}{\partial p} \right], \quad (2)$$

where θ is the potential temperature defined as $\theta = T(p_0/p)^\kappa$, T is temperature, $\vec{V} = (u, v)$ is the horizontal wind vector, ω is the vertical velocity in pressure coordinates, C_p is the specific heat at constant pressure, t is time, p is pressure, $\kappa = R/C_p$ (R is the gas constant for dry air), and $p_0 = 1,000$ hPa. The vertically integrated apparent heat source through the troposphere $\langle Q_1 \rangle$ is calculated as

$$\langle Q_1 \rangle = -\frac{1}{g} \int_{p_s}^{p_T} Q_1 dp, \quad (3)$$

p_s is the surface pressure, and $p_T = 200$ hPa is the pressure at the top of the integration, and g is the gravitational acceleration. The regional average of $\langle Q_1 \rangle$ over the TP above 1.5 km is computed to represent the total diabatic heating of the plateau.

2.6. Wave Activity Flux

The horizontal component of wave activity flux is used to diagnose anomalous wave propagation induced by the mechanical and thermal forcings of the TP (Takaya & Nakamura, 2001). Assuming the Rossby wave is stationary (phase velocity = 0), the formulation in spherical coordinates is as follows:

$$\vec{W} = \frac{p \cos \phi}{2|\vec{V}|} \left[\frac{u}{a^2 \cos^2 \phi} \left[\left(\frac{\partial \psi'}{\partial \lambda} \right)^2 - \psi' \frac{\partial^2 \psi'}{\partial \lambda^2} \right] + \frac{v}{a^2 \cos^2 \phi} \left[\frac{\partial \psi'}{\partial \lambda} \frac{\partial \psi'}{\partial \phi} - \psi' \frac{\partial^2 \psi'}{\partial \lambda \partial \phi} \right] \right] + \frac{u}{a^2 \cos^2 \phi} \left[\frac{\partial \psi'}{\partial \lambda} \frac{\partial \psi'}{\partial \phi} - \psi' \frac{\partial^2 \psi'}{\partial \lambda \partial \phi} \right] + \frac{v}{a^2} \left[\left(\frac{\partial \psi'}{\partial \phi} \right)^2 - \psi' \frac{\partial^2 \psi'}{\partial \phi^2} \right], \quad (4)$$

where ψ' is perturbation quasi-geostrophic stream function, a is Earth radius, and λ and ϕ are longitude and latitude, respectively. In this study, ψ' is computed as the stream function difference between the respective pairs of the CTRL, noTP, and TPnosh experiments.

3. Northern Hemisphere Subtropical High: Climatology and TP Forcing

Figure 1 shows the climatic features of the subtropical high in different seasons, as represented by the 500-hPa stream function (calculated from horizontal winds) and geopotential height fields from the ERA5 reanalysis. The Northern Hemisphere subtropical high is divided into multiple distinct high-pressure centers (Figure 1) due to differences in land and sea distribution and the existence of large-scale topography (W. Li et al., 2012). Seasonally, the subtropical high migrates northward from spring to summer (Figures 1a and 1b) and southward from autumn to winter (Figures 1c and 1d), broadly consistent with the seasonal variation in solar radiation. It reaches its northernmost position and maximum central pressure during summer (Figure 1b).

The multi-model simulations from GMMIP and CESM (Figure S1 in Supporting Information S1) perform well in capturing the observed characteristics of the subtropical high, supporting their use for further analysis of its position and extent. The subtropical high is a dynamically and thermodynamically coupled system, and its features can be effectively captured by the stream function, which describes horizontal flow without relying on the quasi-geostrophic approximation. Both the stream function and geopotential height characterize circulation

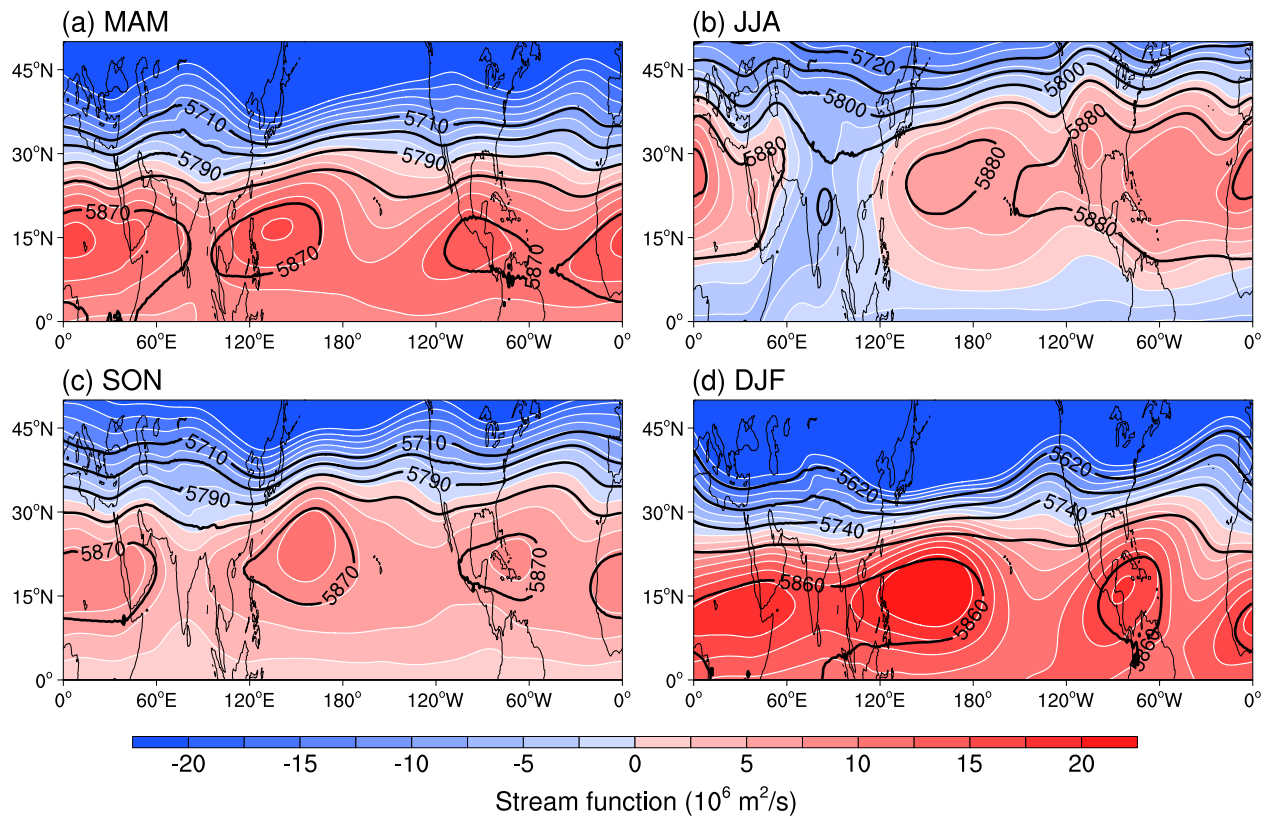


Figure 1. Climatology of the Northern Hemisphere subtropical high during 1979–2014 based on ERA5 reanalysis data. (a) Spring (March–May, MAM), (b) summer (June–August, JJA), (c) autumn (September–November, SON), and (d) winter (December–February, DJF) 500-hPa stream function (shading) and geopotential height (black contours; units: gpm). To highlight differences in spatial patterns across seasons, the basin-mean stream function over 0°N – 50°N has been removed from each panel (subtracted values: 9.0 , 8.0 , 7.6 , and $8.0 \times 10^6 \text{ m}^2 \text{ s}^{-1}$ for panels a–d, respectively).

through their horizontal gradients rather than absolute values. Absolute geopotential height in summer is elevated by thermodynamic atmospheric expansion (Figure 1b), which does not indicate stronger circulation. In fact, both metrics confirm a more intense subtropical high circulation in winter, consistent with stronger winter winds. Previous research has focused on the summer subtropical high not because of stronger circulation but because of its pronounced northward shift, which directly affects Northern Hemisphere continents and contributes to extreme weather events when combined with abundant summer moisture.

To facilitate a clear seasonal comparison of the subtropical high's spatial structure, the basin-mean stream function averaged over 0° – 50°N is removed (Figure 1). This step eliminates the artificial influence of the spherical harmonic boundary condition (which sets the global-mean stream function to zero to ensure a unique solution). The necessity of this removal is evident from Figure S2 in Supporting Information S1: although the circulation patterns differ markedly between summer and winter, their basin-mean values over 0° – 50°N are similar, indicating that regional-mean values are dominated by the artificial boundary condition rather than reflecting genuine seasonal variations in circulation strength. What matters for characterizing the subtropical high is the horizontal gradient (which determines circulation intensity) and the spatial extent of the circulation, neither of which is affected by removing a uniform basin-mean value. As shown in Figure S2 in Supporting Information S1, removing the basin-mean component does not alter the spatial structure or relative intensity (i.e., gradients) of the circulation, but makes seasonal variations easier to analyze. The seasonal migration and structural features revealed by the processed stream function are consistent with the observed climatological evolution of the subtropical high, further validating the use of this diagnostic for quantifying its spatial and seasonal variations.

Figure 2 illustrates the overall effect of the TP on the Northern Hemisphere subtropical high, derived from the difference between the CTRL and noTP experiments (CTRL–noTP). During summer, the TP drives increased stream function and a consequent northward extension and westward expansion of the subtropical high not only

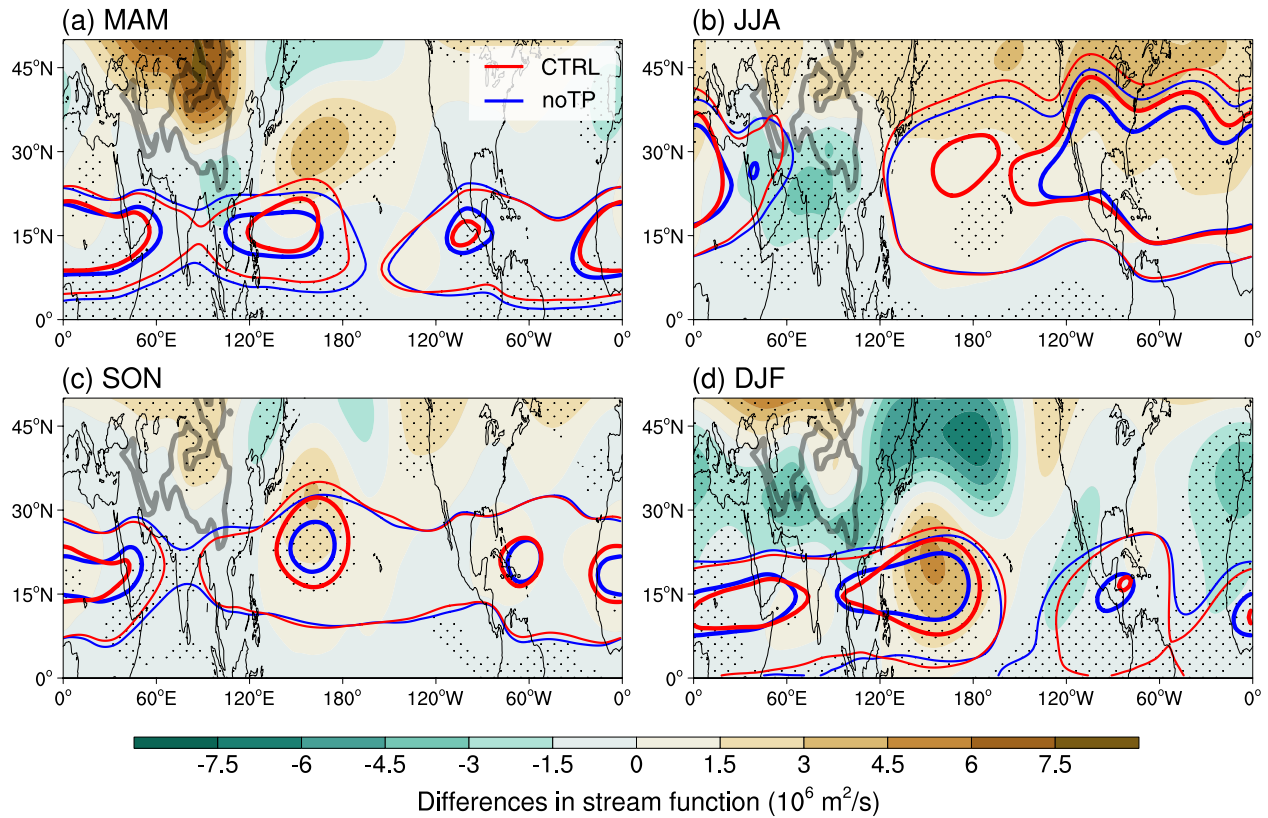


Figure 2. Multi-model mean stream function difference (shading) between the CTRL and noTP experiments (CTRL—noTP) for 1979–2014 represents the overall effect of the TP. Panels show (a) spring (MAM), (b) summer (JJA), (c) autumn (SON), and (d) winter (DJF). The red (CTRL) and blue (noTP) contours trace the same stream function value, outlining changes in the subtropical high, with contour values of 9.0×10^6 and 1.2×10^7 m^2/s in MAM, 0 and 5×10^6 m^2/s in JJA, 4.0×10^6 and 7.5×10^6 m^2/s in SON, and 1.1×10^7 and 1.65×10^7 m^2/s in DJF. Black dots denote areas where the differences are statistically significant at the 95% confidence level and consistent in sign across all three models. The 1.5-km topographic contour is outlined in thick gray.

over the western Pacific but also across the northern North Pacific, North America, and the North Atlantic (Figure 2b). Moreover, the TP drives a northward shift of the WPSH (10°N – 40°N , 110°E – 180°) in spring, and an overall expansion of the WPSH in autumn and winter. Compared to the noTP experiment, the presence of the TP shifts the northern edge of the Northern Hemisphere subtropical high poleward by approximately 1.0° in spring, 2.3° in summer, 1.2° in autumn, and 1.0° in winter (Table 1). Collectively, these findings demonstrate that the TP reorganizes the Northern Hemisphere subtropical high by altering its position and intensity through its combined mechanical and thermal effects.

4. Relative Roles of Mechanical and Thermal Forcing From the TP

The mechanical forcing of the TP is derived from the TPnosh—noTP difference (Figure 3). This difference isolates the mechanical effect because the TPnosh experiment removes the TP thermal forcing while retaining its

Table 1

Contributions of the Overall, Mechanical, and Thermal Effects of the TP to the Hemispheric-Mean Northward Shift of the Northern Hemisphere Subtropical High's Northern Boundary, Diagnosed From the Displacement of Red Contours Relative to Blue Contours

Season	Overall effect (CTRL—noTP)	Mechanical effect (TPnosh—noTP)	Thermal effect (CTRL—TPnosh)
MAM	1.0°	2.5°	-1.5°
JJA	2.3°	-0.9°	3.2°
SON	1.2°	-0.1°	1.3°
DJF	1.0°	1.5°	-0.5°

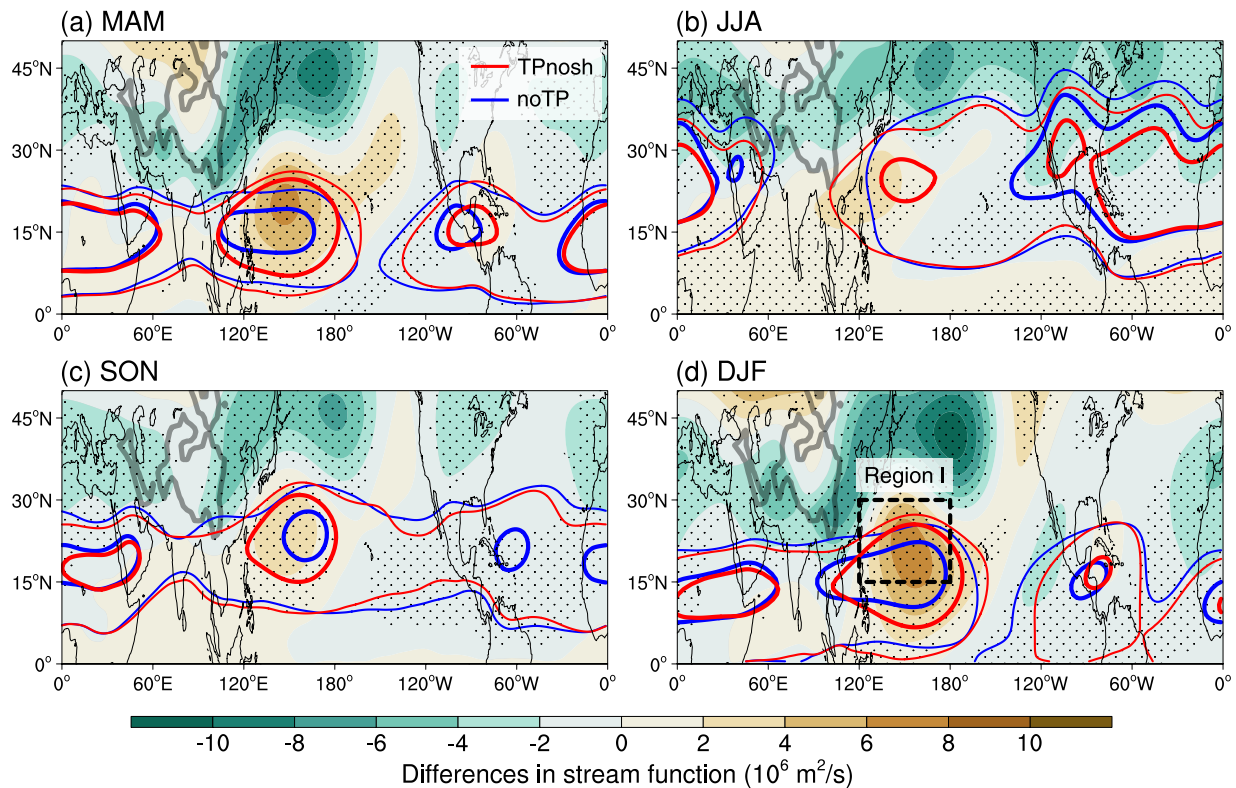


Figure 3. Multi-model mean stream function difference (shading) between the TPnosh and noTP experiments (TPnosh—noTP) for 1979–2014 represents the mechanical effect of the TP. Panels show (a) spring (MAM), (b) summer (JJA), (c) autumn (SON), and (d) winter (DJF). The red (TPnosh) and blue (noTP) contours trace the same stream function value, outlining changes in the subtropical high, with contour values of 9.0×10^6 and 1.2×10^7 m^2/s in MAM, 0 and 5×10^6 m^2/s in JJA, 4.0×10^6 and 7.5×10^6 m^2/s in SON, and 1.1×10^7 and 1.65×10^7 m^2/s in DJF. Black dots denote areas where the differences are statistically significant at the 95% confidence level and consistent in sign across all three models. The dashed rectangle in (d) outlines Region I for subsequent analysis. The 1.5-km topographic contour is outlined in thick gray.

mechanical forcing, whereas the noTP experiment removes both. The mechanical forcing induces the northern edge of the Northern Hemisphere subtropical high northward by approximately 2.5° in spring, -0.9° in summer, -0.1° in autumn, and 1.5° in winter (Table 1). This forcing induces a pronounced dipole pattern in the stream function downstream of the TP, strongest in spring and winter. In spring and winter, the mechanical effect drives negative stream function anomalies over the equatorial eastern Pacific, shifting the American subtropical high (15°N – 45°N , 80°W – 20°W) eastward (Figures 3a and 3d). In contrast, during summer and autumn, the widespread mid-latitude negative anomalies lead to a southward retreat of the Northern Hemisphere subtropical high, excluding the WPSH (Figures 3b and 3c). The WPSH, located near the positive pole of this dipole, exhibits an overall expansion in spring and winter (Figures 3a and 3d), little change in its central position, and expands westward in summer and autumn (Figures 3b and 3c).

The thermal effect of the TP (CTRL—TPnosh) is distinct from the mechanical effect at the hemispheric scale. Thermal forcing shifts the northern edge of the Northern Hemisphere subtropical high northward by approximately -1.5° in spring, 3.2° in summer, 1.3° in autumn, and -0.5° in winter (Table 1), exhibiting an inverse pattern to the mechanical effect. For example, the mechanical forcing produces a pronounced downstream dipole pattern (positive in the south, negative in the north; Figure 3), whereas the thermal forcing generates a reversed dipole pattern (negative in the south, positive in the north; Figure 4). Furthermore, the widespread mid-latitude negative anomalies seen in summer and autumn under mechanical forcing (Figures 3b and 3c) change to positive anomalies under thermal forcing (Figures 4b and 4c). The thermal forcing of the TP causes the WPSH to weaken in spring and winter and retreat eastward in summer and autumn (Figure 4). Although the two effects partially offset each other, the net influence of the TP is clearly evident in the overall response (Figure 2). Mechanically induced strengthening of the downstream WPSH dominates in winter, whereas the thermal effect prevails in summer, driving a northward expansion and intensification of the Northern Hemisphere subtropical high.

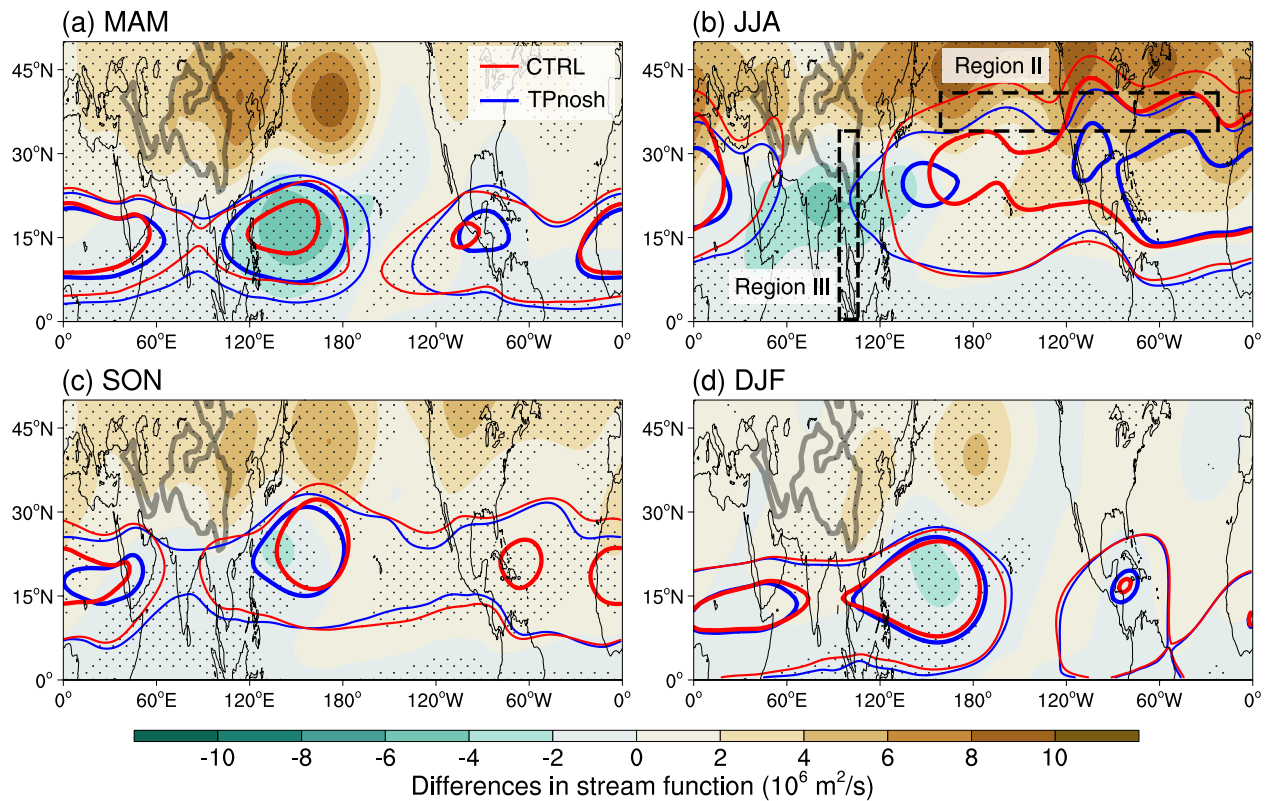


Figure 4. Multi-model mean stream function difference (shading) between the CTRL and TPnosh experiments (CTRL—TPnosh) for 1979–2014 represents the thermal effect of the TP. Panels show (a) spring (MAM), (b) summer (JJA), (c) autumn (SON), and (d) winter (DJF). The red (CTRL) and blue (TPnosh) contours trace the same stream function value, outlining changes in the subtropical high, with contour values of 9.0×10^6 and 1.2×10^7 m^2/s in MAM, 0 and 5×10^6 m^2/s in JJA, 4.0×10^6 and 7.5×10^6 m^2/s in SON, and 1.1×10^7 and 1.65×10^7 m^2/s in DJF. Black dots denote areas where the differences are statistically significant at the 95% confidence level and consistent in sign across all three models. The dashed rectangles in (b) outline Regions II and III for subsequent analysis. The 1.5-km topographic contour is outlined in thick gray.

A composite analysis based on reanalysis data was conducted to compare with the model sensitivity experiments. Years with summer TP heat source ($\langle Q_1 \rangle$) anomalies exceeding one standard deviation were selected from 1940 to 2022, yielding 10 high and 11 low thermal forcing years (Figure 5a). This analysis focuses on summer, when

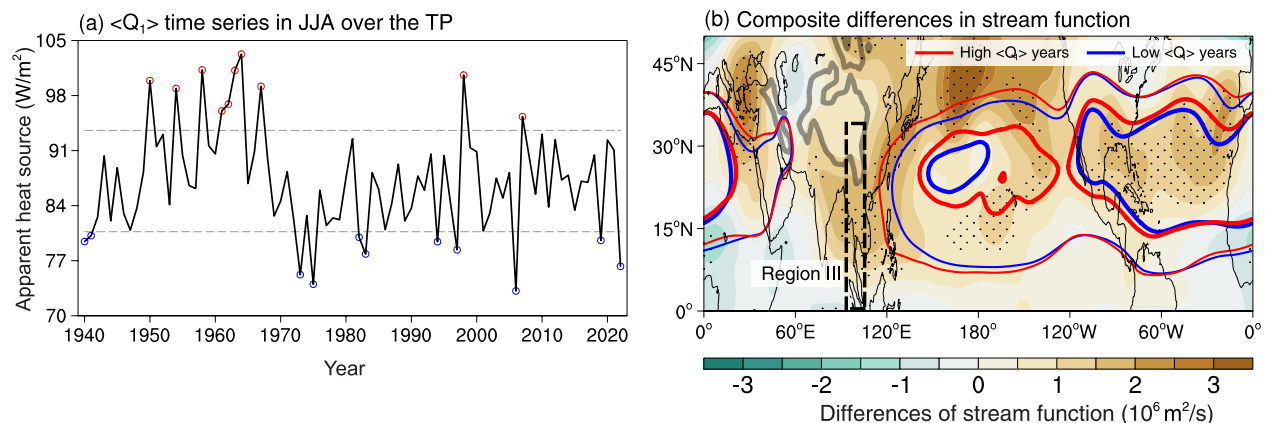


Figure 5. Composite analysis of the TP thermal forcing effect on the Northern Hemisphere subtropical high in summer (JJA; 1940–2022) based on ERA5. (a) Time series of the vertically integrated apparent heat source ($\langle Q_1 \rangle$; Equation 3) averaged over the TP region above 1.5 km (thick gray contour in (b)). The red and blue circles mark the high and low thermal forcing years defined by the ± 1 standard deviation threshold (gray dashed lines). (b) Stream function difference (shading) between high and low thermal forcing years. The red and blue contours represent characteristic streamlines for high and low forcing years, respectively, outlining changes in the subtropical high. Black dots indicate significance at the 95% confidence level. The dashed rectangle in (b) outlines Region III for subsequent analysis.

TP thermal forcing exerts its strongest influence on the atmosphere (G. Wu et al., 2007; Xie, Wu, et al., 2023). The composite results are broadly consistent with those from the sensitivity experiments. Specifically, high thermal forcing years exhibit significant increases in stream function across the Northern Hemisphere mid-latitudes compared with low forcing years (Figure 5b), indicating a northward expansion and intensification of the Northern Hemisphere subtropical high.

The stream function anomalies in the observational composite are smaller in magnitude than those in the modeled thermal effect (Figure 5b), resulting in a relatively weaker northward expansion of the subtropical high. This difference is expected because interannual variations in TP heating in observations represent a more moderate forcing compared to the complete removal of sensible heating in the TPnosh experiment. Despite this difference in magnitude, both the observations and the model simulations agree on the large-scale response of the Northern Hemisphere subtropical high in summer. A key discrepancy, however, lies in the response of the WPSH. In contrast to the thermal effect simulated by the model (Figure 4b), the observations show positive stream function anomalies south of the TP and a westward extension of the WPSH (Figure 5b). This pattern likely reflects the influence of air-sea interactions over the Indian Ocean (B. He et al., 2019; Z. Wang et al., 2019), a process not captured by the atmosphere-only model experiments.

5. Mechanisms of TP Influence on the Northern Hemisphere Subtropical High

Having established the distinct impacts of the TP mechanical and thermal forcings on the Northern Hemisphere subtropical high, this section investigates the underlying dynamical processes. Based on the finding that mechanical forcing dominates in winter and thermal forcing in summer, the analysis focuses on three key regions: Region I (15°–30°N, 120°–180°E; Figure 3d), located on the northern flank of the WPSH under winter mechanical forcing; Region II (35°–40°N, 160°E–20°W; Figure 4b), situated at the boundary zone of the subtropical high influenced by summer thermal forcing; and Region III (0°–35°N, 95°–105°E; Figures 4b and 5b), covering the southern TP and adjacent areas affected by summer thermal forcing. Regions I, II, and III are selected to represent the strongest and most spatially coherent circulation responses to TP mechanical and thermal forcings, as identified from the anomaly patterns in Figures 3d and 4b. Besides, minor changes in the regional boundaries do not affect the inferred mechanisms (Figures S3–S5 in Supporting Information S1). The remote influence of the TP on the Northern Hemisphere subtropical high is investigated by examining geopotential height, wave activity flux, vertical motion, meridional circulation, and precipitation to reveal the roles of atmospheric Rossby waves, westerly jet stream, and vertical circulation cells.

In winter, Region I exemplifies the intensification and northward shift of the WPSH driven by TP mechanical forcing, characterized by vertically coherent positive geopotential height anomalies that intensify upward and peak in the mid-to-upper troposphere (Figures 6a and 6b). These positive anomalies result from a barotropic Rossby wave train triggered by the TP topography (Figure 6c). Although topographically forced Rossby waves mainly propagate downstream, the TP can also affect its adjacent upstream circulation (Wu, 1984). This upstream response arises because the TP mechanically deflects the background westerly flow, directly altering the vorticity and pressure fields immediately upstream of the plateau, rather than resulting from any independent wave source in the upstream region. Consequently, while the wave activity flux remains directed downstream (Figure 6c), the resulting wave train establishes geopotential height anomalies on both sides of the TP, with notably stronger anomalies downstream, modulating the intensity and position of the westerly jet. Figure 6c confirms that the wave train downstream of the TP is dominant, while the upstream wave train corresponds to the TP's influence on its adjacent upstream circulation, consistent with results from the theoretical Rossby wave model (Lutsko & Held, 2016). This wave train propagates southeastward from the TP to the western Pacific, generating positive stream function anomalies over Region I and negative anomalies to its northeast, which together establish the dipole pattern seen in the overall and mechanical effects of the TP (Figures 2d and 3d).

In summer, Region II exemplifies the northward expansion of the Northern Hemisphere subtropical high driven by the TP thermal forcing, characterized by a belt of positive geopotential height anomalies at 500 hPa between 30°N and 60°N (Figure 7b), which is barotropic throughout the troposphere (Figure 7a). This anomaly belt reduces the geopotential height gradients to the south while enhancing them to the north. As shown in Figure 7e, weakened westerlies on the equatorial flank and strengthened westerlies on the poleward side are observed, jointly facilitating the northward expansion of the subtropical high.

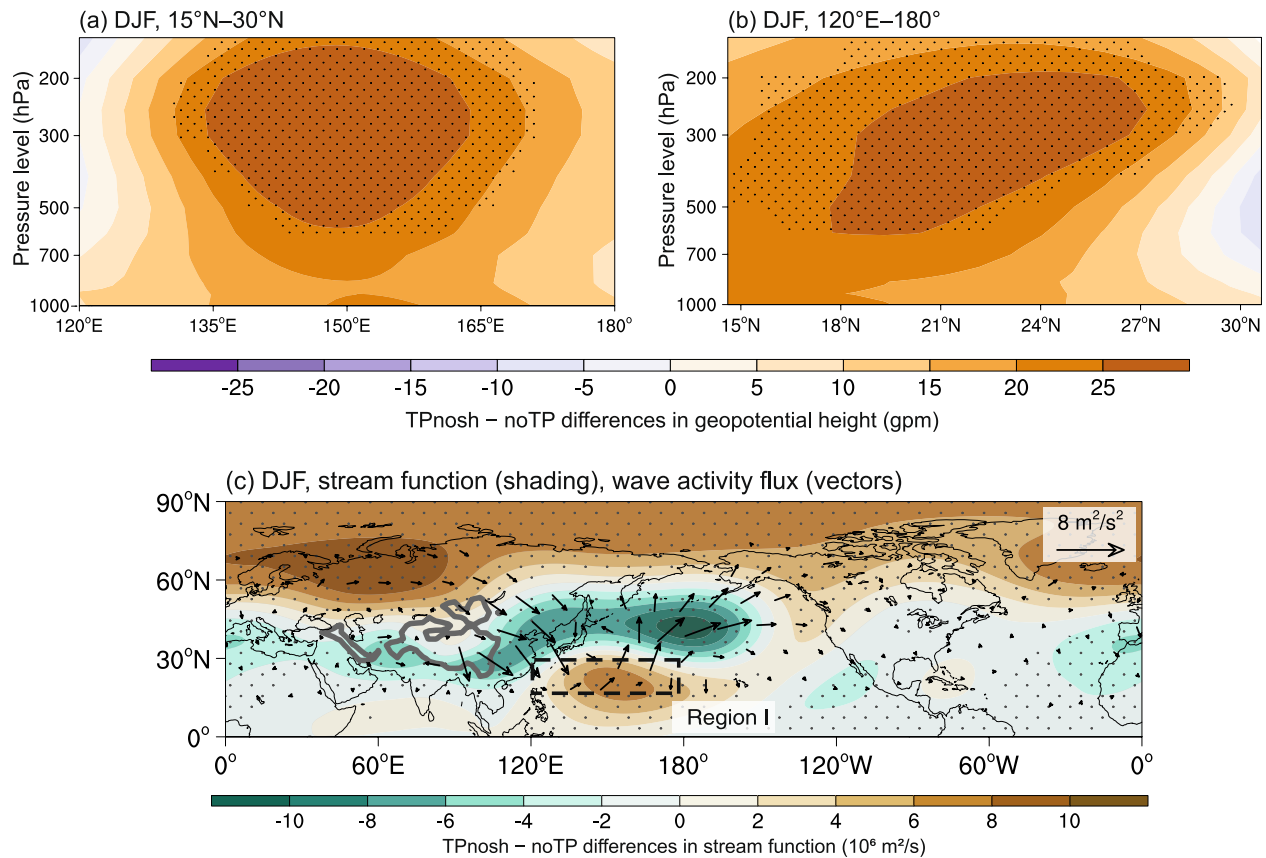


Figure 6. Multi-model mean differences (TPnosh—noTP) in Region I (15°–30°N, 120°–180°E; Figure 3d) during winter (DJF). (a) Zonal-mean (15°–30°N) and (b) meridional-mean (120°–180°E) vertical profiles of geopotential height. (c) 500-hPa stream function difference (shading) and wave activity flux (vectors; Equation 4). The 1.5-km topographic contour is shown in thick gray. Black dots denote areas where the differences are statistically significant at the 95% confidence level and consistent in sign across all three models. The dashed rectangle in (c) outlines Region I.

As indicated by the wave activity flux, TP thermal forcing stimulates an anomalous Rossby wave train that propagates along the belt of positive geopotential height anomalies (Figure 7c). This wave train produces alternating anticyclonic and cyclonic anomalies within the positive belt (Figure 7b). Owing to the barotropic structure of the Rossby wave and mass continuity, these horizontal circulation anomalies induce a corresponding vertical motion pattern, with subsidence displaced to the east of the positive anomalies and ascent near the negative anomalies (Hoskins et al., 2003). This near-quadrature relationship favors the development and maintenance of the wave structure.

Region III exemplifies the southward influence of TP thermal forcing, a feature evident in both model simulations and observations, though notable differences exist in the detailed response (Figures 4b and 5b). The simulated thermal forcing induces a baroclinic response over the southern slopes of the TP in summer, marked by lower-tropospheric cyclonic and upper-tropospheric anticyclonic anomalies (Figure 8a). This vertical structure stems from intense diabatic heating and strong ascent (Figure 8b), which drive a meridional–vertical circulation characterized by ascent over the southeastern TP and Indochina, accompanied by subsidence farther south (G. Wu et al., 2016). The 25°–35°N region exhibits lower-tropospheric convergence and upper-tropospheric divergence, inducing upward motion. In contrast, the 0°–15°N region shows the opposite configuration, resulting in subsidence (Figure 8b). The resulting enhanced rainfall and latent heating sustain the summer monsoon, while suppressed precipitation occurs south of 15°N (Figure 8c).

In contrast to the simulated eastward retreat of the WPSH under TP thermal forcing, reanalysis data show a westward extension. Dynamically, the observed ascending branch, upper-tropospheric divergence coupled with lower-tropospheric convergence and negative geopotential height center are located farther north than in the simulations, accompanied by a poleward shift in the precipitation anomaly (Figure 9). This discrepancy indicates

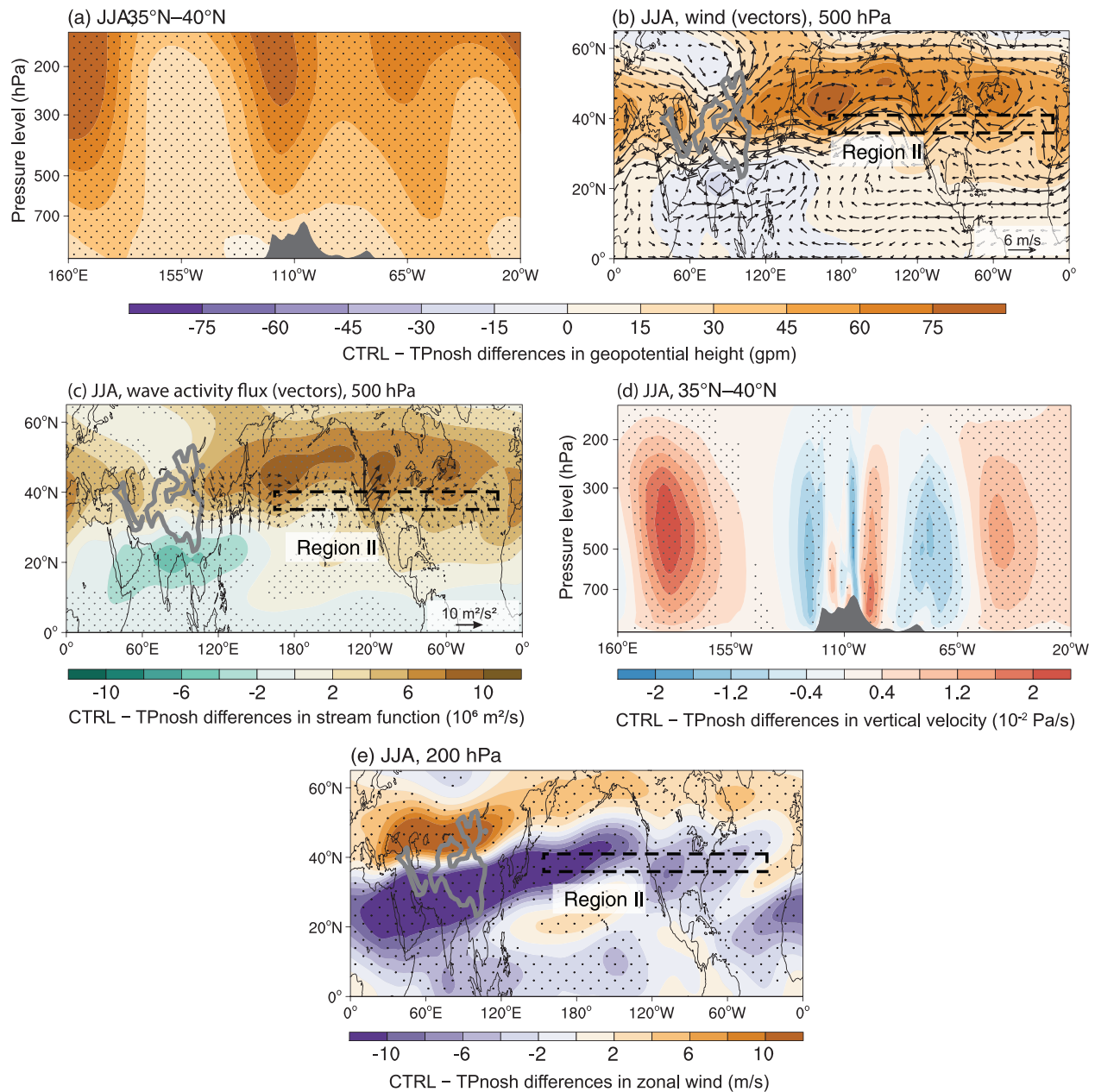


Figure 7. Multi-model mean differences (CTRL–TPnosh) in Region II (35°–40°N, 160°E–20°W; Figure 4b) during summer (JJA). (a) Zonal-mean (35°–40°N) vertical profile of geopotential height. (b) 500-hPa geopotential height difference (shading) and horizontal wind (vectors). (c) 500-hPa stream function difference (shading) and wave activity flux (vectors; Equation 4). (d) Zonal-mean (35°–40°N) vertical profile of vertical velocity. (e) 200-hPa zonal wind (shading). The 1.5-km topographic contour is shown in thick gray. The dashed rectangle in (b, c, e) outlines Region II. Black dots denote areas where the differences are statistically significant at the 95% confidence level and consistent in sign across all three models.

that additional processes beyond the direct atmospheric response to TP thermal forcing are involved. A key factor is the absence of air–sea interactions in the atmosphere-only model experiments. Previous studies have shown that TP thermal forcing can induce significant sea surface temperature anomalies in the tropical Indian Ocean through changes in surface heat fluxes and large-scale circulation (B. He et al., 2019; Z. Wang et al., 2019). For example, air–sea feedbacks over the tropical Indian Ocean can enhance the sensitivity of the WPSH to orographic forcing (Kitoh, 2004) while partially offsetting the TP's direct thermal effects (B. He et al., 2019; Z. Wang et al., 2019) collectively produce a compensating response that alters the large-scale circulation pattern. This coupled

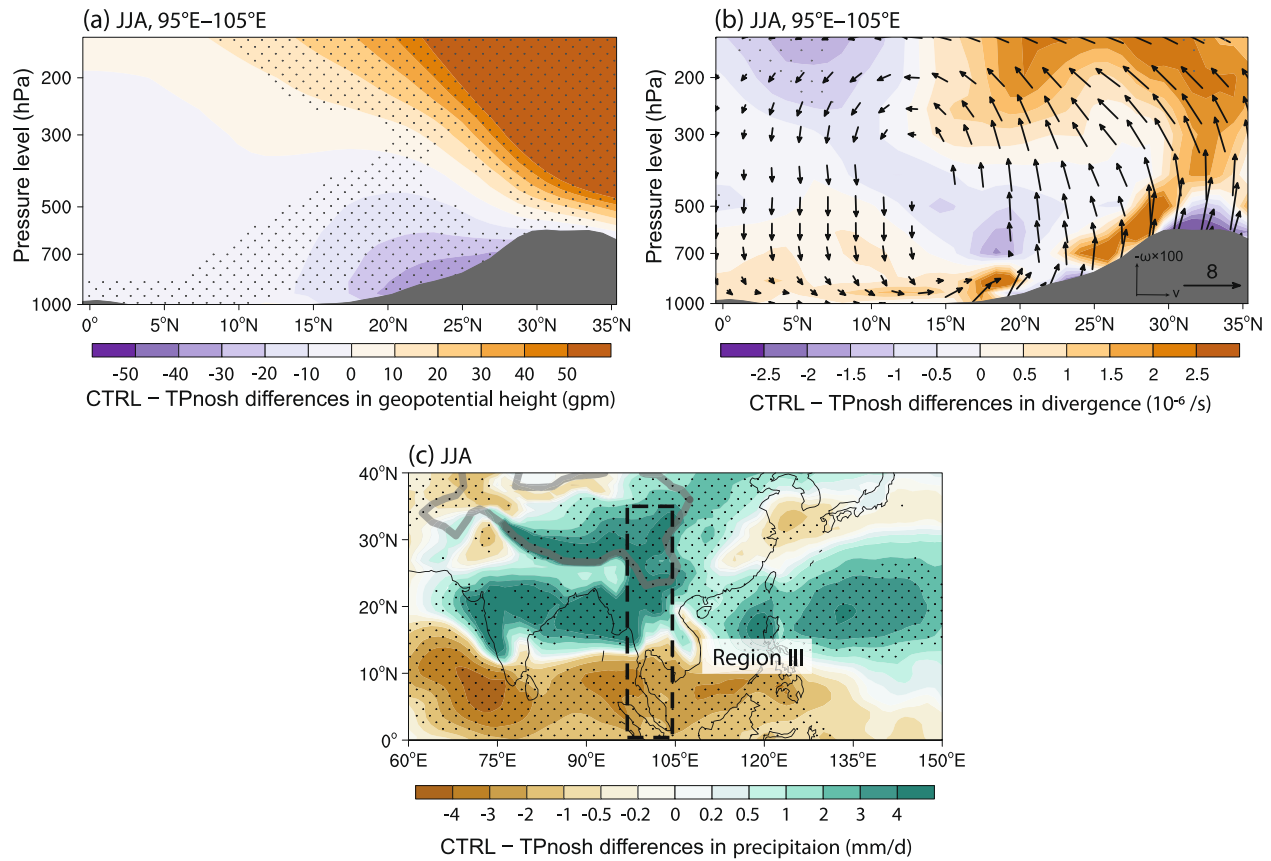


Figure 8. Multi-model mean differences (CTRL—TPnosh) in Region III (0° – 35° N, 95° – 105° E; Figure 4b) during summer (JJA). (a) Meridional-mean (95° – 105° E) vertical profile of geopotential height. (b) Meridional-mean (95° – 105° E) vertical profile of divergence (shading) and meridional-vertical circulation (vectors). (c) Precipitation difference. The 1.5-km topographic contour is shown in thick gray. The dashed rectangle in (c) outlines Region III. Black dots denote areas where the differences are statistically significant at the 95% confidence level and consistent in sign across all three models. Vectors in (b) are plotted only where both components are statistically significant and consistent in sign across models. Gray shading in (a) and (b) indicates topography.

adjustment favors a poleward shift of convection and weakens the low-latitude response, contributing to the northward-displaced anomalies and the westward extension of the WPSH seen in observations.

Moreover, a diagnosis of the heat source reveals that this discrepancy is also driven by modeled and observed differences in diabatic heating distribution. The spatial pattern of the heat source mirrors the precipitation anomalies (Figures 8c and 9c, and Figures S6a and S6c in Supporting Information S1), indicating a crucial role of latent heating release associated with precipitation. Vertically, the simulated heating over the TP in the CTRL—TPnosh difference is concentrated in the near-surface layer (Figure S6b in Supporting Information S1), reflecting the dominant role of surface sensible heating in the model's thermal forcing. In contrast, the ERA5 composite differences (high minus low thermal forcing years) show a heating peak at higher levels (around 700 hPa; Figure S6d in Supporting Information S1), indicating that latent heating from precipitation plays a more prominent role in observations. These distinct vertical structures of diabatic heating between the model and observations can lead to different atmospheric circulation responses (C. He et al., 2019; H. Luo et al., 2024), which in turn modulate precipitation and its associated latent heating feedback, ultimately contributing to the observed differences in the subtropical high response.

6. Conclusions and Discussion

Our findings demonstrate that the TP shapes the Northern Hemisphere subtropical high through distinct seasonal mechanisms. In addition to intensifying and expanding the WPSH overall via mechanical forcing in winter, its intense summer thermal forcing induces a more extensive reorganization, driving a hemispheric-scale northward expansion of the subtropical high.

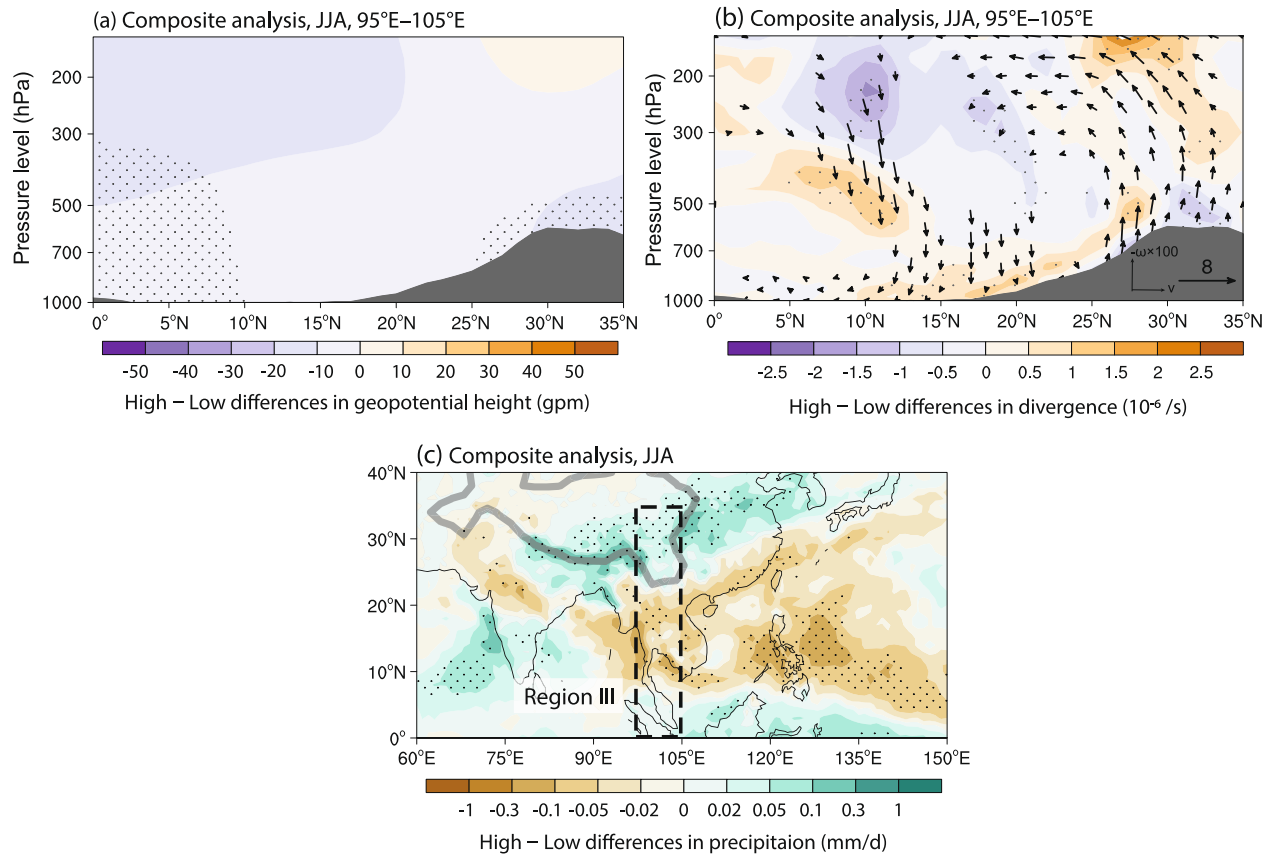


Figure 9. Differences between high and low thermal forcing years in Region III (0°–35°N, 95°–105°E; Figure 5b) during summer (JJA) based on the ERA5 data set. (a) Meridional-mean (95°–105°E) vertical profile of geopotential height differences. (b) Meridional-mean (95°–105°E) vertical profile of divergence differences (shading) and meridional-vertical circulation (vectors). (c) Precipitation difference. The 1.5-km topographic contour is shown in thick gray. The dashed rectangle in (c) outlines Region III. Black dots denote areas where the differences are statistically significant at the 95% confidence level. Vectors in (b) are plotted only where both components are statistically significant. Gray shading in (a) and (b) indicates topography.

Dynamically, the winter mechanical forcing triggers a downstream Rossby wave train, which establishes a deep high-pressure anomaly over the western Pacific, leading to an intensified and overall-expanded WPSH (Figure 10a). In contrast, the summer thermal forcing generates a zonal belt of high-pressure anomalies and associated Rossby wave trains, resulting in a hemispheric-scale northward expansion of the subtropical high. Consistent with the barotropic Rossby wave structure and mass continuity, the amplification and maintenance of the wave structure are accompanied by a near-quadrature vertical motion pattern (Figure 10b). Observations indicate that this summer forcing also induces a meridional-vertical circulation to the south of the TP, characterized by ascent over the plateau and descent to its south, which drives a westward extension of the WPSH. The simulated eastward retreat of the WPSH, conversely, likely stems from the lack of tropical air-sea interactions in the model experiments and a different diabatic heating distribution between models and observations.

While previous research has predominantly examined the thermal influence of the TP on the summertime WPSH (Duan et al., 2017), a systematic quantification of its mechanical and overall impacts on the Northern Hemisphere subtropical high has been lacking. This study provides a comprehensive assessment of the TP's role across the entire hemisphere. By strategically separating the mechanical and thermal forcings in multi-model simulations, we quantify their distinct and seasonally varying contributions, thereby revealing a complete picture of how the TP shapes these pivotal circulation systems.

Model and observation discrepancies in the zonal displacement of the WPSH point to inadequate representation of tropical air-sea coupling in our experiments (B. He et al., 2019; Z. Wang et al., 2019). Future studies should therefore utilize fully coupled models and targeted experiments to unravel the synergistic effects of the TP and

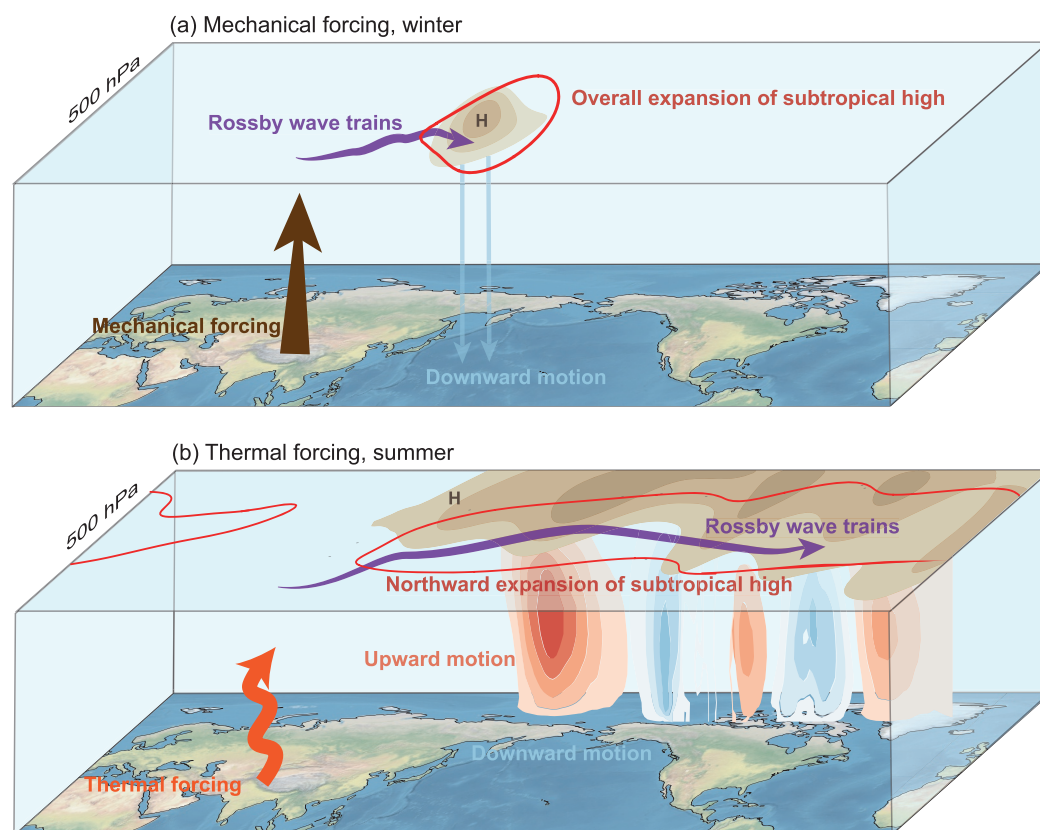


Figure 10. Schematic diagram illustrating how the large-scale atmospheric circulation associated with the TP shapes the Northern Hemisphere subtropical high. (a) Winter mechanical forcing triggers a downstream Rossby wave train and deep high-pressure anomaly, intensifying and expanding the WPSH. (b) Summer thermal forcing generates wave trains, an associated zonal high-pressure belt and a near-quadrature vertical motion pattern, driving a hemispheric northward expansion of the Northern Hemisphere subtropical high.

tropical oceans on the subtropical high. Our findings establish the TP thermal forcing as a pivotal driver of the Northern Hemisphere subtropical high in summer, which governs the distribution of extreme weather. This mechanistic link implies that antecedent TP signals must be incorporated into hemispheric extreme weather risk assessment, modulating events such as heatwaves (M. Luo & Lau, 2017), heavy precipitation (Polade et al., 2017), and marine heatwaves (Yao & Wang, 2021). These signals offer a tangible pathway to improve the forecasting of hydroclimatic extremes, as shown with spring soil temperature anomalies (Q. Li et al., 2025), thereby enhancing societal preparedness.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

ERA5 data can be accessed at <https://doi.org/10.24381/cds.adbb2d47>, <http://doi.org/10.24381/cds.f17050d7>, <http://doi.org/10.24381/cds.6860a573> and <https://doi.org/10.24381/cds.bd0915c6>. GMMIP data are available at <https://esgf-node.llnl.gov/search/cmip6/>. Data from the CESM simulations and codes are available at <https://zenodo.org/doi/10.5281/zenodo.17538770>.

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